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Ener-Habitat: An online numerical tool to evaluate the thermal performance of homogeneous and non-homogeneous envelope walls/roofs

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Abstract

The online numerical tool Ener-Habitat, designed to evaluate the thermal performance of homogeneous and non-homogeneous walls or roofs of the building envelope using a time-dependent heat transfer model, is presented. This tool is useful in evaluating constructive systems composed by up to seven homogeneous layers and also constructive systems composed by one non-homogeneous layer plus up to six homogeneous layers. The non-homogeneous layer can be either filled or hollow block type. Ener-Habitat solves the time-dependent heat transfer equation with convective boundary conditions on both sides, in one-dimension for homogeneous constructive systems and in two-dimensions for constructive systems containing one non-homogeneous layer. It takes into consideration that the constructive system is either in a non air-conditioned room or in an air-conditioned one. The one-dimensional model for homogeneous multilayered constructive systems is validated by comparison with EnergyPlus results. The two-dimensional model is validated by calculating the heat transfer through a concrete hollow block wall in steady-state and the comparison with experimental results. Ener-Habitat is proved to be a reliable tool.

Keywords: Thermal performance; Simulation tool; Time dependent; Building envelope; Validation; Online tool

1. Introduction

Over the past 50 years, hundreds of building thermal performance programs have been developed (Crawley et al., 2008). A directory of building energy software tools can be found in U.S. Department of Energy (2014a). These software tools can be classified in two types: whole-building and single component. There are some that consider the time-dependent heat transfer that occurs in the real situation, but others consider a steady-state or quasi-steady-state disregarding the effect of the thermal capacity (Passive House Institute, 2014; National Renewable Energy Laboratory, 2014). This last assumption is a good approximation only when used for air-conditioned buildings during winter in temperate or polar zones (Kuehn et al., 2001). In this work, the term air-conditioned building means that the indoor air temperature is kept constant.

The dynamic whole-building type thermal simulation tools use the time-dependent heat transfer model for each component of the whole building, They can provide users with key building performance indicators in real situations, but they require a very specific and arduous training process, and knowledge in heat transfer, to be properly used. There are programs that can perform simulations for both air or non air-conditioned buildings (U.S. Department of

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Energy, 2014b; Solar Energy Laboratory, 2014; Free Software Foundation, Inc., 2014; Danish Building Research Institute, 2014; Autodesk, 2014; Nöel, 2014a), and only for air-conditioned buildings (U.S. Department of Energy, 2014c). Whole-building type simulation tools present some limitations in performing simulations where there are air cavities within walls or roofs (Kim and Park, 2011).

The thermal performance single component type software tools consider that the only heat transfer to the room is through the envelope building component (wall, roof or window) under evaluation. This kind of programs are useful to compare different constructive systems for a component. Most of these programs were developed for a specific research, for envelope wall or roof constructive systems composed by homogeneous layers (Kossecka and Kosny, 2002; Vijayalakshmi et al., 2006; Ozel and Pihtili, 2007; Barrios et al., 2011; Al-Sanea et al., 2012; Al-Sanea et al., 2013) and for constructive systems composed by homogeneous layers and one non-homogeneous layer (Gao et al., 2004; Al-Hazmy, 2006; Al-Khameis et al., 2011; Zhang et al., 2014). Also programs of this type have been developed to be used by building designers, consultants, teachers and students. In all of them, the constructive system is considered to be in an air-conditioned room, with a fixed indoor air temperature, and almost all of them consider a steady-state condition. Construction *R*-value Calculator (Design Navigator, 2014) is an online tool to calculate the *R*-value of a large number of common wall or roof constructive systems. It uses the isothermal planes method to account for thermal bridging, considering a steady-state. Therm (Lawrence Berkeley National Laboratory, 2014) and AnTherm (Tomasz Komicki, 2014) are computer programs developed especially to evaluate non-homogeneous layered components, where thermal bridges are of concern, in a steady-state. These programs model two-dimensional conduction heat transfer in building components, AnTherm can also model in three-dimensions. Convection and radiation inside the air-cavities are approximated through the use of an effective conductivity. KaLiBat (Nöel, 2014b) calculates the linear heat loss coefficient of two-dimensional thermal bridges in non-homogeneous constructive systems without air-cavities, considering a steady-state. Heat2 (Buildingphysics.com, 2014a) and Heat3 (Buildingphysics.com, 2014b) respectively, calculate two-dimensional and three-dimensional heat conduction within constructive systems that can be described in a rectangular grid, for both transient and steady-state. These programs can import climatic data for dynamic calculations. Convection and radiation inside the air-cavities are approximated through the use of an effective conductivity. Lesokai (Eggimann et al., 2000) calculates the overall heat transfer coefficient for homogeneous layered constructive systems. Although thermal bridges are detected, this program does not calculate them. The calculations are made using the methods of the Swiss and European regulations under steady-state and harmonic boundary conditions. Opaque (2014) was designed to calculate the *U*-value, the decrement factor, and the time lag for homogeneous and non-homogeneous constructive systems using time-dependent climatic conditions, nevertheless it does not calculate correctly the decrement factor and the time lag.

Ener-Habitat is an online numerical tool developed with the aim of providing Mexican building designers, teachers and students, with an easy-to-use tool that can help them select an adequate constructive system for envelope walls and roofs in Mexican climates (Huelsz et al., 2013). Ener-Habitat is a single component type software tool. The main characteristics that made it unique are the use of a time-dependent heat transfer model; the possibility to evaluate either systems composed by homogeneous layers (using a one-dimensional model) or by homogeneous layers plus one non-homogenous layer (using a two-dimensional model), including constructive systems with air-cavities; the possibility to evaluate the wall or roof either used in an air conditioned room or in a non air-conditioned one; and the online free access (www.enerhabitat.unam.mx). The second-to-last characteristic is an important issue due to the fact that a constructive system has a different thermal performance depending on the condition of the room where it is used (Barrios et al., 2011). Ener-Habitat can help in the development of new Mexican regulations, as current standards (Secretaría de Energía, 2001; Secretaría de Energía, 2011) and evaluation systems (INFONAVIT, 2014) use the steady-state heat transfer model. Ener-Habitat can also be a useful tool for countries with similar climates. The possibility to extend the cities database is open, especially for Spanish-speaking countries, as Ener-Habitat is written in this language.

In this paper, the tool Ener-Habitat is presented and validated. Section 2 presents the tool Ener-Habitat, including the heat transfer models used for both types of constructive systems, the numerical method used to solve the models, the construction of the climate data, and a description of its use procedure. The validations of Ener-Habitat heat transfer models, for the homogeneous and for non-homogeneous constructive systems, are presented in Sections 3 and 4, respectively. Concluding remarks are pointed out in Section 5.

2. Ener-Habitat

Ener-Habitat is a thermal performance single component type software tool, thus it considers that the only heat transfer to the room is through the wall or roof under evaluation. In Ener-Habitat (EH) the thermal evaluation of a wall or roof constructive system (CS) is made for the typical day of each month, considering periodic conditions. EH is capable of evaluating the thermal performance of a CS composed only by homogeneous layers (HCS) or composed by homogeneous layers plus one non-homogeneous layer (NHCS). EH has a database with the most used materials in Mexico and their thermal properties. Also, each user has a user database where can define his own materials with their corresponding thermal properties.

A description of the heat transfer models for both types of constructive systems, the numerical method used to solve the models, the construction of the climate data of EH, and the procedure of its use, are presented in the following subsections.

2.1. Heat transfer model for homogeneous constructive systems

The one-dimensional time-dependent heat transfer model is used for HCS. This model was already presented in Barrios et al. (2011) and was validated with the analytical solution of a one-dimensional layer problem, in which, one side of the layer is initially at a given temperature and suddenly the temperature is changed to a different constant value, while the other side of the layer is adiabatic (Patankar, 1980).

The heat transfer equation of the one-dimensional time-dependent heat transfer model, for each layer of a system with N homogeneous layers, is (Incropera et al., 2007)

$$\rho_j c_j \frac{\partial T_j}{\partial t} = k_j \frac{\partial^2 T_j}{\partial x^2}, \tag{1}$$

where T_j is the temperature, ρ_j, c_j, and k_j are the density, the specific heat, and the thermal conductivity, respectively of the $j-th$ layer, $j = 1, \ldots, N$. Also t is the time and x the coordinate perpendicular to the surface of the constructive system.

By continuity of the heat transfer at the layers joints,

$$k_j \left.\frac{\partial T}{\partial x}\right|_{j,j+1} = k_{j+1} \left.\frac{\partial T}{\partial x}\right|_{j,j+1}. \tag{2}$$

At the outdoor surface, $x = 0$, the boundary condition is

$$h_o (T_{sa} - T_{x=0}) = -k \left.\frac{\partial T_1}{\partial x}\right|_{x=o}, \tag{3}$$

where h_o is the outdoor film coefficient, $T_{x=0}$ is the outdoor surface temperature, and T_{sa} is the sol–air temperature defined as (ASHRAE, 1997)

$$T_{sa} = T_o + \frac{I_s a}{h_o} - RF. \tag{4}$$

In this expression, T_o is the outdoor air temperature, I_s is the solar radiation over the wall or roof, a is the outdoor surface solar absorptance, and RF is the infrared radiation factor for the wall or roof. In EH $h_o = 13\ \mathrm{W/m^2\,K}$ (Secretaría de Energía, 2011; Secretaría de Energía, 2001).

At the indoor surface, $x = L$, L being the constructive system total width, the boundary condition is

$$-k \left.\frac{\partial T_N}{\partial x}\right|_{x=L} = h_i (T_{x=L} - T_i), \tag{5}$$

where h_i is the indoor film coefficient, $T_{x=L}$ is the indoor surface temperature and T_i is the indoor air temperature. In EH, for a wall $h_i = 8.1\ \mathrm{W/m^2\,K}$ and for a roof, if the heat flows from outdoor to indoor $h_i = 9.4\ \mathrm{W/m^2}\,K$, otherwise $h_i = 6.6\ \mathrm{W/m^2\,K}$. The values are taken from Mexican regulations (Secretaría de Energía, 2011; Secretaría de Energía, 2001).

When considering an air-conditioned room, the indoor air temperature is kept constant and set at the neutrality temperature defined by (Humphreys and Nicol, 2000)

$$T_n = 0.54\overline{T_o} + 13.5\ ^\circ\mathrm{C}, \tag{6}$$

where $\overline{T_o}$ is the month average outdoor air temperature.

When a non air-conditioned room is considered, the indoor air temperature is a function of the heat transmitted through the wall or roof, and is calculated by

$$H \rho_a c_a \frac{dT_i}{dt} = h_i (T_{x=L} - T_i), \tag{7}$$

where H is the distance where an adiabatic boundary condition is assumed, and ρ_a and c_a are the density and the specific heat of the air, respectively. In EH $H = 2.5\ m$.

2.2. Heat transfer model for non-homogeneous constructive systems

For NHCS, EH uses the two-dimensional time-dependent heat transfer model. As pointed out previously, two types of non-homogeneous layers can be solved by EH, filled block or hollow block types, shown in Fig. 1. The filled block type has a solid frame with cavities filled by another solid material, while in the hollow block type the cavities of the solid frame have air. The materials can be selected from EH database or the user database.

Two-dimensional heat transfer by conduction is modeled by the two-dimensional extension of Eq. (1), with the boundary conditions in the x direction, similar to Eqs. (3) and (5), and symmetry conditions in the y direction ($\partial T/\partial y = 0$). In NHCS filled block type, heat conduction is the only heat transfer mechanism involved. Continuity of the heat flux is used in the boundaries between the two solids, similar to Eq. (2). In NHCS hollow block type, three heat transfer mechanisms are involved: conduction through the solid material, natural convection of the air inside the cavities, and radiation between cavity surfaces. Natural convection is taken into account by using a temperature dependent heat transfer coefficient appropriate to the cavity geometry (Xamán et al., 2005; Hollands et al., 1975) and boundary conditions between air and interior surfaces similar to Eq. (5). The heat transmitted between cavity interior surfaces by radiation is simulated using the assumptions that the air is a non-participant fluid, the cavity interior surfaces are diffuse and gray, and the emitted thermal radiation by a surface corresponds to a blackbody at its average temperature. To calculate the radiation received from the other surfaces the corresponding view factors are used (Incropera et al., 2007). The air cavity temperature is a function of the heat transmitted between the air and the cavity surface, which is calculated using a similar equation

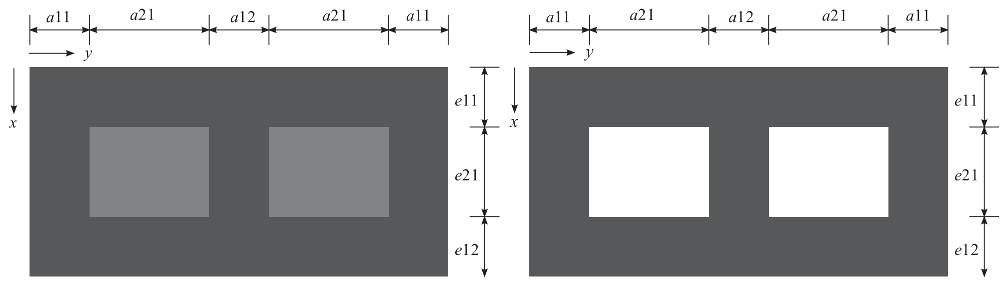

Fig. 1. Non-homogeneous layer types of non-homogeneous constructive systems. The filled block type (left) is composed by two solid materials and the hollow block type (right) has air cavities. The dimensions $a11$, $a21$, $a12$, $e11$, $e21$ and $e12$ should be specified by the user.

to Eq. (7), but considering the four surfaces of the cavity and replacing H by the area of the cavity.

As for the one-dimensional model, when considering an air-conditioned room, the indoor air temperature is kept constant at the neutrality temperature given by Eq. (6), and when a non air-conditioned room is considered, the indoor air temperature is calculated by Eq. (7).

2.3. Numerical method

For HCS and NHCS, the corresponding time-dependent heat transfer equation is solved numerically by using an implicit control volume scheme. For the HCS, the set of N_x equations are solved via a Tridiagonal Matrix Algorithm (TDMA) for each time step, N_x being the number of control volumes used. The harmonic mean conductivity procedure is used to calculate the effective thermal conductivity for each face of the control volumes (Patankar, 1980). EH uses a time step of $1s$ and $N_x = 100$. Each time the TDMA is applied, an exact solution is found for the set of N_x equations. For the NHCS, a set of $N_x \times N_y$ equations are solved, N_y is the number of control volumes in the perpendicular direction to x. EH uses a time step of $1s$ and $N_x = N_y = 160$. For NHCS, an iterative process is implemented with a convergence criterion for each time step

$$|T_{l,m}(t) - T_{l,m}(t')| \leqslant C_{TDMA}, \text{ for } l = 1, \ldots, N_x, \quad m = 1, \ldots, N_y, \tag{8}$$

where $T_{l,m}(t)$ and $T_{l,m}(t')$ are the old and new temperatures, respectively, inside the TDMA iteration, l and m are the position of the control volume in the x and y direction, respectively, and C_{TDMA} is the convergence parameter. In EH $C_{TDMA} = 1 \times 10^{-10}$ °C.

Because the solution must be periodic, for both the one and the two-dimensional heat transfer models, the following criterion is used (here specified for the two-dimensional model)

$$|T_{l,m}(0h) - T_{l,m}(24h)| \leqslant C_S, \text{ for } l = 1, \ldots, N_x, \quad m = 1 \ldots, N_y, \tag{9}$$

where $T_{l,m}(0h)$ and $T_{l,m}(24h)$ are the temperature at the beginning of the day and the temperature at the end of the day and $C_S = 1 \times 10^{-5}$ °C.

2.4. Climate data construction

EH has a climate database of 80 cities of Mexico that includes the latitude and data for the typical day of each month. For each city and each month, the database contains the minimum and the maximum of the outdoor air temperature, the time when the maximum temperature occurs, and the maxima of the beam and the diffuse solar radiations on a horizontal surface. These data were obtained averaging the meteorological data of the month (Meteonorm, 2014). The outdoor air temperature as a function of time is generated from the minimum and maximum temperatures of the day and the time they occur, using the model proposed by Chow and Levermore (2007). The time corresponding to the minimum temperature is assumed to happen at sunrise. The beam and diffuse solar radiations on a horizontal surface are assumed to be the positive part of a sine function and are calculated with the sunrise and sunset as starting and ending times, the amplitude being its corresponding maximum value. The solar radiation on the wall or roof, I_s, in Eq. (4), is calculated as the sum of the beam and diffuse solar radiations on the corresponding surface. Sunrise, sunset, and the position of the sun for the beam radiation calculation, are calculated for the 15th day of the month (Duffie and Beckman, 1980). The diffuse solar radiation, I_d, is assumed to vary linearly with the wall or roof inclination angle in degrees, θ_β, $I_d = I_{dh}(180 - \theta_\beta)/180$, where I_{dh} is the diffuse solar radiation on a horizontal surface. The infrared radiation factor, RF, is also assumed to vary linearly with the inclination of the wall or roof, considering 3.9 °C for a horizontal surface and zero for a vertical one (ASHRAE, 1997).

2.5. Procedure of use

The procedure to use the tool EH is simple. First, the user selects the type of CS to evaluate (HCS or NHCS). The user then chooses the city from a list of 80 Mexican cities, specifies the simulation period (the month or the whole year for HCS, the month for NHCS), sets the condition inside the room (non air-conditioned or air-conditioned), and defines the location of the CS (wall or roof). For HCS, the user must set the number of CS to evaluate (1–5). For NHCS, the user can evaluate only one CS at a time. Next, the user sets the inclination and orientation of the CS. Then, the user defines each CS using up to seven layers. For HCS, the user sets the thickness and the material of each layer, the user must also specify the outdoor surface solar absorptance of the outdoor layer. For the NHCS, the user first selects the type of non-homogeneous layer (filled block or hollow block) and sets the dimensions and the materials. Then, the user defines one by one up to six homogeneous layers, in the same way that for the HCS, and selects the position of the non-homogeneous layer within the system. EH has a database with the most commonly used materials and their corresponding thermal properties. It is possible for the user to define new materials within a user database. Once the CS have been defined the user can start the numerical simulations.

When the simulations finish, EH displays the results page, which is different depending on the condition inside the room, non air-conditioned or air-conditioned.

For CS in a non air-conditioned room, EH presents the decrement factor and allows to download a file with this and other seven evaluation parameters. The file contains the decrement factor of the indoor air temperature with respect to the sol–air temperature and the corresponding lag time, the energy transmitted through the CS, the average indoor air temperature, the maximum and minimum indoor air temperatures, the hot and cold thermal performance indexes, and the discomfort degree hour (Barrios et al., 2012). In another file, the user can download the time evolution of the outdoor air temperature, indoor air temperature, indoor surface temperature, and the sol–air temperature. In the case of annual simulations (only for HCS), EH reports the results for the typical day of each month and the annual average of each evaluation parameter.

For CS in an air-conditioned room, EH presents the cooling, heating, and total energy per unit area per day required to keep the indoor air temperature at the neutrality temperature. In the downloadable file, these parameters, and the surface decrement factor are given. In the case of annual simulations (only for HCS), EH presents the cooling, heating, and total energy corresponding to the typical day of each month and the annual value of each parameter.

3. Validation of the model for homogeneous constructive systems

In this work the one-dimensional model for homogeneous constructive systems is validated by comparing EH and EnergyPlus (E+) results. In this section, the methodology to recreate similar conditions to EH in E+, a description of the HCS study cases, and the results for non air-conditioned and for air-conditioned rooms are presented.

3.1. Methodology

A methodology to use E+ to individually evaluate a CS of a single envelope component, as EH does, is proposed here. The first step is to create a cubic module with no windows and doors. The evaluated CS must have the desired orientation. The non evaluated envelope components, including the floor, should be defined with no mass, adiabatic, and with no external radiation exchange. To define no mass, the material is created in the *Material: NoMass* object (U.S. Department of Energy, 2014b). In this object, the only thermal property that appears is the *Thermal Resistance*. This property can be set at any value, since it will be defined as adiabatic. To define the surfaces as adiabatic, the option *Adiabatic* is selected for the *Outside Boundary Condition* in the *BuildingSurface: Detailed* object. To set the no external radiation exchange condition, the *Thermal Absorptance* of the formerly created material in the *Material: NoMass* object must be set close to zero, since it cannot be zero (in this case study it is set as 1×10^{-5}). Afterwards, a *Construction* is created containing the material with no mass and emissivity close to zero (1×10^{-5}). This *Construction* is selected in the *BuildingSurface: Detailed* object for all non evaluating envelope components.

To compare results from EH an E+ programs, the same conditions of EH are set in E+. Thus, a cubic module of 2.5 m length is used, without any internal heat gain, and the infiltration and ventilation are set to zero. The sol–air temperature corresponding to the evaluating HCS in the typical day of the month, obtained from EH, is used to create an EPW file using the Weather and Statistics Converter program. This weather file has the sol–air temperature in the place of the air temperature and all solar radiation components are set to zero, and repeats all year long in the EPW. Because the effect on the evaluated CS of the infrared radiation is included in the sol–air temperature, the emissivity of the evaluated CS is set close to zero (1×10^{-5}). In E+, the values for the outdoor and indoor film coefficients, h_o and h_i, are set equal to that of EH through the *Surface Convection Heat Transfer Coefficient EMS Actuator*.

3.2. Study cases

Simulations utilizing EH and E+ were carried out for six HCS in the East wall, for non air-conditioned rooms and for air-conditioned rooms. The simulations were made for the hot month, May, in Torreón, Coahuila, Mexico. Table 1 shows the description of the HCS used and the thermal properties of the materials are listed in Table 2. In E+, the heat transfer through the envelope components is solved using the conduction transfer functions with 60 steps per hour.

3.3. Results

For the comparison between EH and E+ in a non air-conditioned room, the parameters are the indoor air temperature, T_i, the decrement factor, DF, of the indoor air temperature with respect to the sol–air temperature, and

Table 1
Description of the homogeneous constructive systems (HCS). Layers are given from outdoor to indoor with thickness shown in parentheses. Materials: expanded polystyrene (EPS), high-density concrete (HDC), lightweight plaster (LP), mortar and brick. All HCS have an outdoor surface solar absorptance of 0.4.

HCS	Description from outdoor to indoor layers
1	(10 cm) EPS
2	(10 cm) HDC
3	(2 cm) EPS + (8 cm) HDC
4	(8 cm) HDC + (2 cm) EPS
5	(1 cm) Mortar + (14 cm) Brick + (1 cm) LP
6	(1 cm) Mortar + (2.54 cm) EPS + (14 cm) Brick + (1 cm) LP

Table 2
Thermal properties of the different materials used in the homogeneous constructive systems: thermal conductivity, k, density, ρ, and specific heat, c. Materials: expanded polystyrene (EPS), high-density concrete (HDC), lightweight plaster (LP), brick, and mortar.

Material	k (W/m °C)	ρ (kg/m^3)	c (J/kg °C)
EPS	0.04	15	1400
HDC	2.00	2400	1000
LP	0.16	1000	600
Brick	0.70	1970	800
Mortar	0.88	2800	896

the corresponding lag time, LT. Fig. 2 presents T_i as a function of time. It can be noticed that the results obtained from EH are in good agreement with those of E+. The largest difference is 0.5 °C for HCS 6, the one with the smallest oscillation amplitude. The average of T_i for HCS is equal for all HCS, 32.2 ± 0.2 °C, since they have the same solar absorptance. In Fig. 3 the DF and LT are presented. As can be seen, the results from EH and E+ are in good agreement, and both programs give the same order of the HCS from best to worst in terms of thermal performance, given by the DF. The largest difference of DF between EH and E+ results is 0.03, for HCS 1. The largest difference of LT is 0.4 h, for HCS 4, HCS 5, and HCS 6.

When evaluating the HCS for the air-conditioned room, the parameters used are the cooling energy, E_c, and the heating energy, E_h, per unit area per day. The results obtained from EH and E+ are in good agreement, as can be seen in Fig. 4. Both EH and E+ give the same order of HCS from best to worst in terms of thermal performance for cooling and for heating. The largest difference between EH and E+ in E_c, is 6.68 W h/m^2 day (3%), for HCS 5. For E_h, the largest difference is 4.64 W h/m^2 day (4%), for HCS 2.

4. Validation of the model for the non-homogeneous constructive systems

The two-dimensional model for NHCS for the filled block type was validated by setting the same material for the frame and the cavities, and comparing results from the one-dimensional model. The results are the same. For the NHCS hollow block type, the two-dimensional model is validated by comparing the results of the model used by EH with experimental results in the steady-state. This validation is presented in this section. It is worth to note that no experimental nor numerical results have been found in the literature in a time-dependent state, either for the filled block type nor for the hollow block type.

4.1. Methodology

The numerical code of EH that solves the two-dimensional time-dependent heat transfer model for NHCS was used to solve the steady-state heat transfer through a hollow block type. The results of the heat transmitted through the wall per unit area, from the numerical code are compared with experimental results for a concrete hollow block wall of 2.4 m^2 area reported in Borbón et al. (2010) and Pérez et al. (2011). The hollow blocks have dimensions $a11 = 0.024$ m, $a21 = 0.16$ m, $a12 = 0.048$ m, $e11 = e12 = 0.025$, and $e21 = 0.07$ m (see Fig. 1(b)). The concrete of the hollow block has thermal conductivity $k_c = 1.1$ W/m °C and emissivity $\epsilon = 0.93$. The experiments were carried out using a hot-box type set-up, where the temperature at the outdoor, $T_{x=0}$, and indoor, $T_{x=L}$, surfaces were set constant. The heat flux was measured for different values of $T_{x=0}$ and $T_{x=L}$. In order to be able to compare the results from the EH two-dimensional code and the experiments, the sol–air temperature, T_{sa}, and indoor air temperature, T_i, in EH, are set equal to the values of $T_{x=0}$ and $T_{x=L}$, respectively, and the values for the indoor and outdoor film coefficients are set to $h_i = h_o = 1000$ W/m^2 K. Then, in EH the values at $T_{x=0}$ and $T_{x=L}$ are equal to the experimental values, without modifying the EH code. In order to reach the steady-state faster, the temperature initial condition of all control volumes is set equal to the average of $T_{x=0}$ and $T_{x=L}$.

4.2. Results

The heat flux through the wall per unit area calculated by EH, q_{EH}, is compared with the experimental value, q_{exp}, reported in Borbón et al. (2010), for different values of $T_{x=0}$ and $T_{x=L}$, the results are shown in Table 3. The experimental uncertainty of q_{exp} is $\pm 3.8\%$. As can be observed, EH results are in good agreement with the experimental ones, the maximum difference is 6.5%. In general EH slightly undervalues the heat transmitted and only in case 6, EH overvalues it.

5. Concluding remarks

In this work, Ener-Habitat, an online numerical tool for the evaluation of the thermal performance of envelope constructive systems for walls or roofs using time-dependent heat transfer, is presented and validated. This tool has the capability to evaluate the constructive systems when used in non air-conditioned and in air-conditioned rooms.

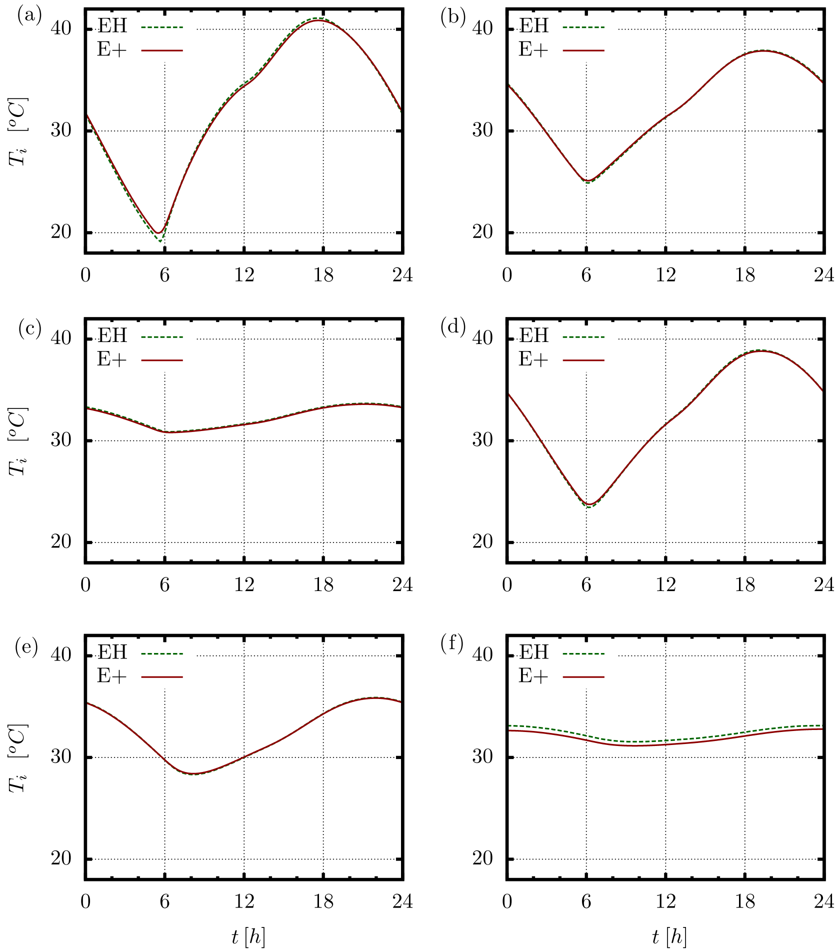

Fig. 2. Indoor air temperature, T_i, as a function of time, t, obtained from Ener-Habitat (EH) and from EnergyPlus (E+), for the six HCS: (a) 1, (b) 2, (c) 3, (d) 4, (e) 5, and (f) 6.

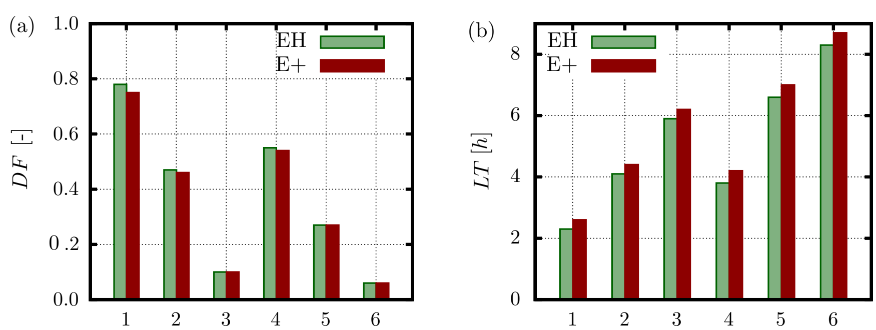

Fig. 3. (a) Decrement factor, DF, and (b) lag time, LT, for the six HCS obtained from Ener-Habitat (EH) and EnergyPlus (E+).

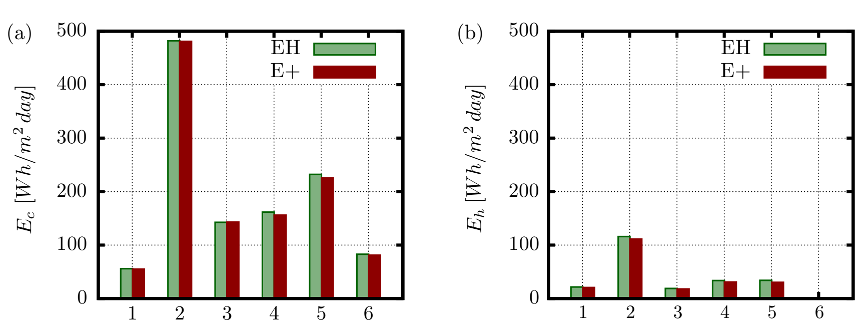

Fig. 4. (a) Cooling energy, E_c, and (b) heating energy, E_h, for the six HCS obtained from Ener-Habitat (EH) and EnergyPlus (E+).

Table 3
The heat flux through the wall per unit area, experimental value, q_{exp}, taken from Borbón et al. (2010) and the calculated by EH, q_{EH}, for different values of $T_{x=0}$, $T_{x=L}$, and $\Delta T = T_{x=L} - T_{x=0}$. Percentage difference between EH and experimental results, e.

Case	$T_{x=0}$ (°C)	$T_{x=L}$ (°C)	ΔT (°C)	q_{exp} (W/m^2)	q_{EH} (W/m^2)	e (%)
1	42.4	28.8	13.6	78.03	75.45	3.3
2	41.6	24.7	16.9	93.53	93.11	0.4
3	40.7	20.7	20.0	112.48	109.72	2.4
4	40.1	16.8	23.3	135.50	126.62	6.5
5	59.4	32.4	26.9	163.12	158.61	2.7
6	63.5	33.3	30.2	171.27	179.99	5.1
7	61.6	21.7	39.8	238.07	232.16	2.5

The constructive systems can be either composed by up to seven homogeneous layers or by one non-homogeneous layer plus up to six homogeneous layers. The non-homogeneous layer can be either filled block type or hollow block type.

The validation of the one-dimensional model for homogeneous constructive systems is given by the good agreement with EnergyPlus results. For non air-conditioned rooms, the maximum difference in temperature is 0.5 °C, in the DF 0.03, and in the LT 0.4 h. For air-conditioned rooms, the maximum difference in E_c is 3% and in E_h is 4%.

The validation of the two-dimensional model for non-homogeneous constructive systems is carried out by a comparison with the one-dimensional model for the filled block type. The results are the same. The validation for the hollow block type is done by comparing results with experiment results of the heat flux in steady-state through a concrete hollow block wall. The maximum difference is 6.5%.

It can be concluded that Ener-Habitat is a reliable tool for evaluating the thermal performance of a building single envelope component, wall or roof, either for homogeneous or non-homogeneous constructive systems, in non and air-conditioned rooms. It has the advantage of its easy use and that it does not require that the user has knowledge in heat transfer. Ener-Habitat can be utilized to evaluate the thermal performance of constructive systems for countries with similar climatic conditions to the ones found in Mexico. In the present version of Ener-Habitat (V2.2), the non-homogeneous layers available are limited to two configurations for walls, and two configurations for roofs. Future versions will include more non-homogeneous layer configurations.

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